United States Fatal Traffic Accident Analysis

Marshall Nicholson

April 22, 2026

Research Questions

- **Is a drunk driver more likely to kill themselves or another driver?**

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- **Is a drunk driver more likely to kill themselves or another driver?**

Context

Why does this matter?

- Importance of regulations

Figure: NHTSA Logo

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Data Source:

- National Highway Traffic Safety Administration

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Limitations:

- Hard to replace NANs
- 3 missing values

Figure: NHTSA Logo

Methods and Analysis

Data cleaning techniques used:

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- **File type switch**

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- **Row removal**

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- File type switch
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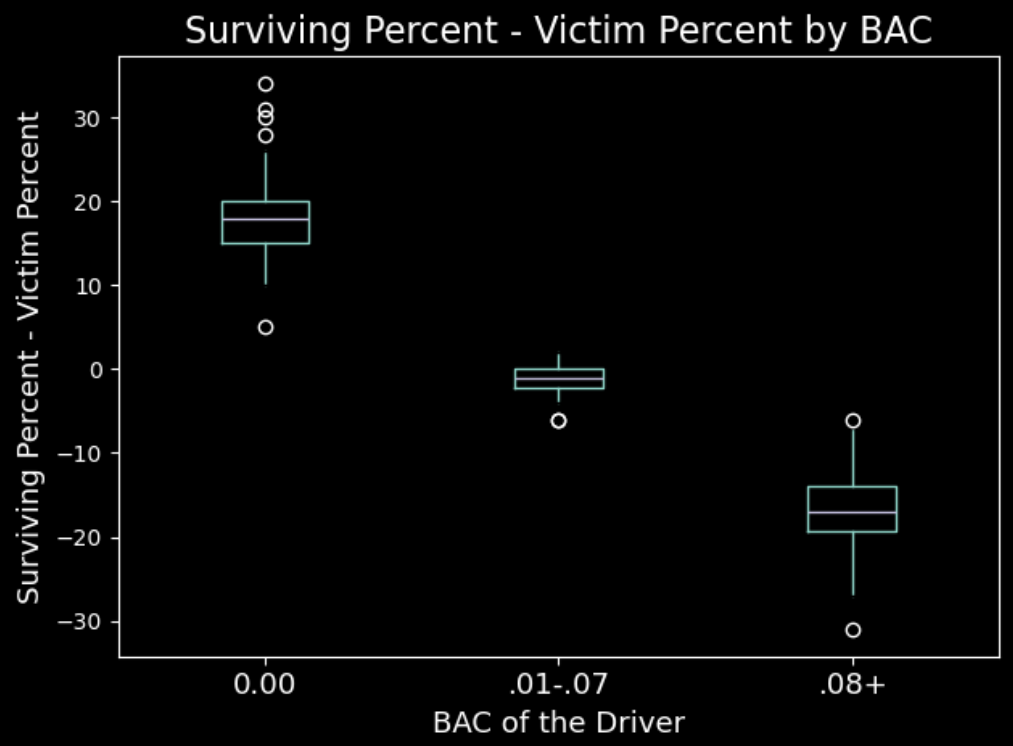
- Row removal

My data columns are not independent, making the statistical tests we learned invalid. Therefore much of the analysis is based on visuals.

Driver BAC Survival Rate in Fatal Crashes

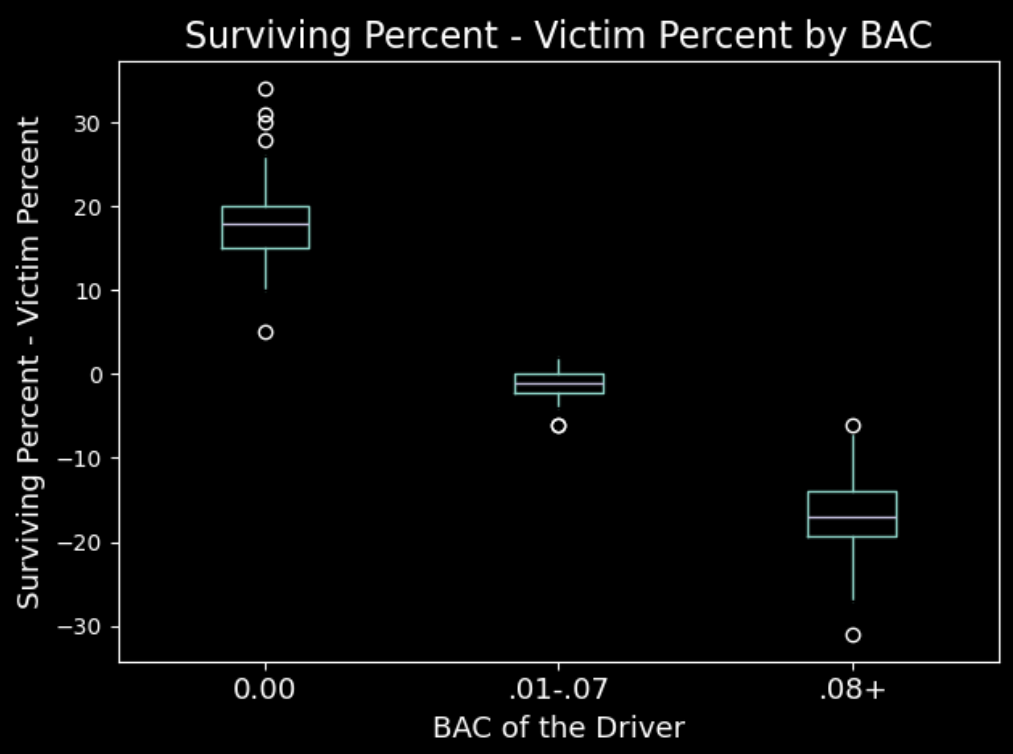
In a fatal crash:



Driver BAC Survival Rate in Fatal Crashes

In a fatal crash:

- **20% more drunk driver died than survived**



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In a fatal crash:

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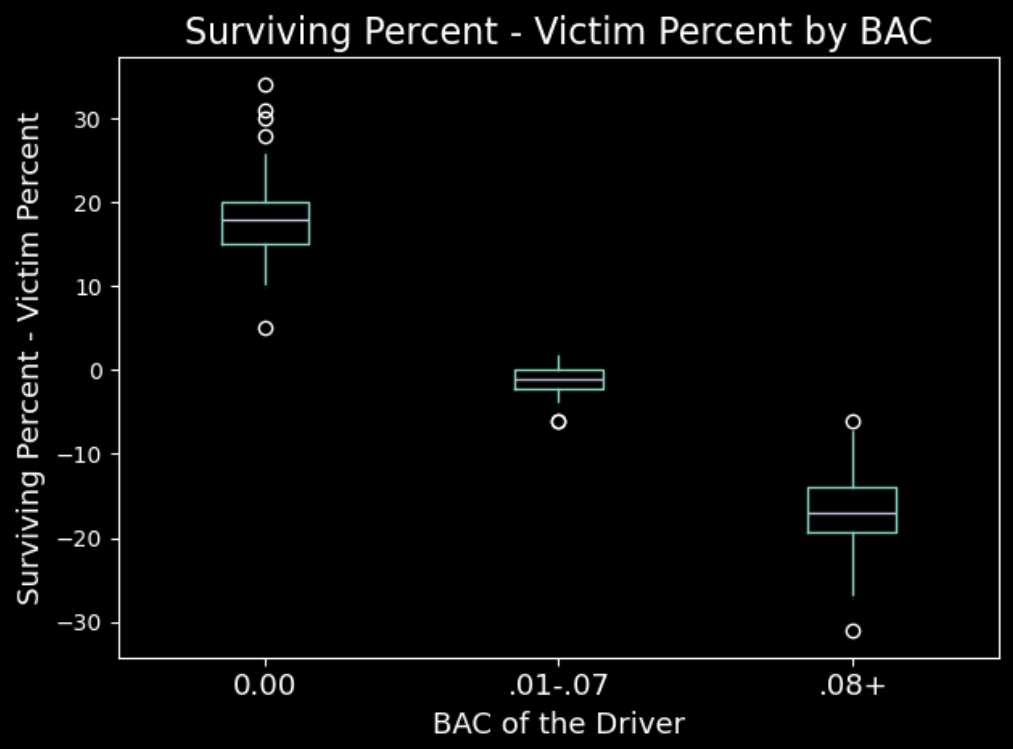
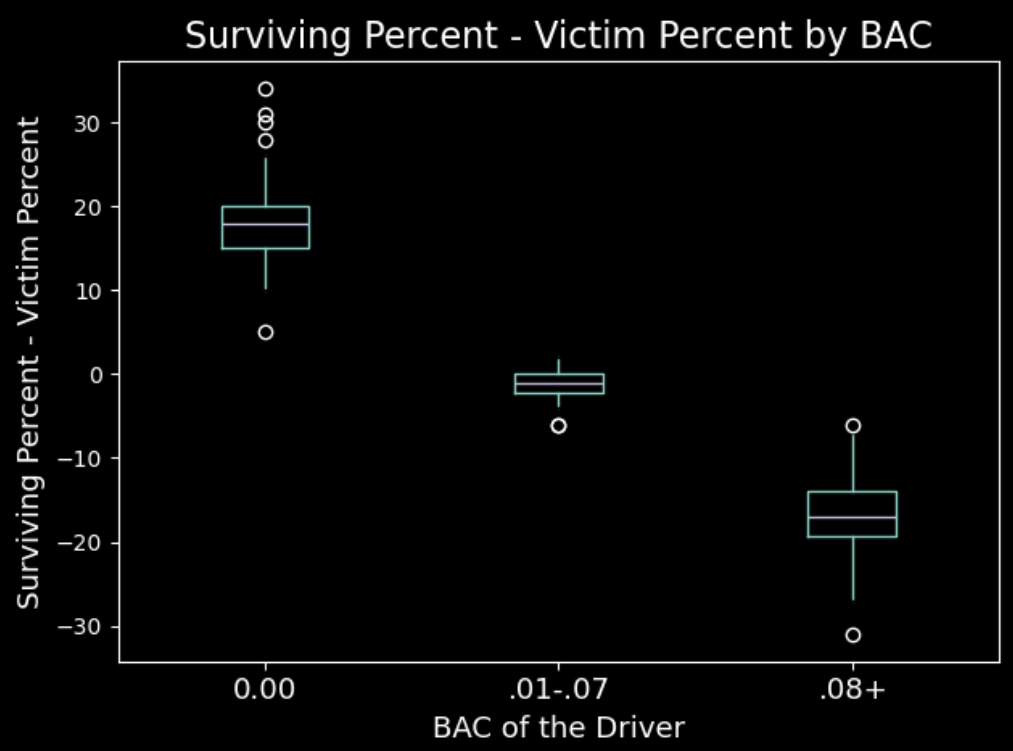
- **In some states, the percentage is much worse**



Driver BAC Survival Rate in Fatal Crashes

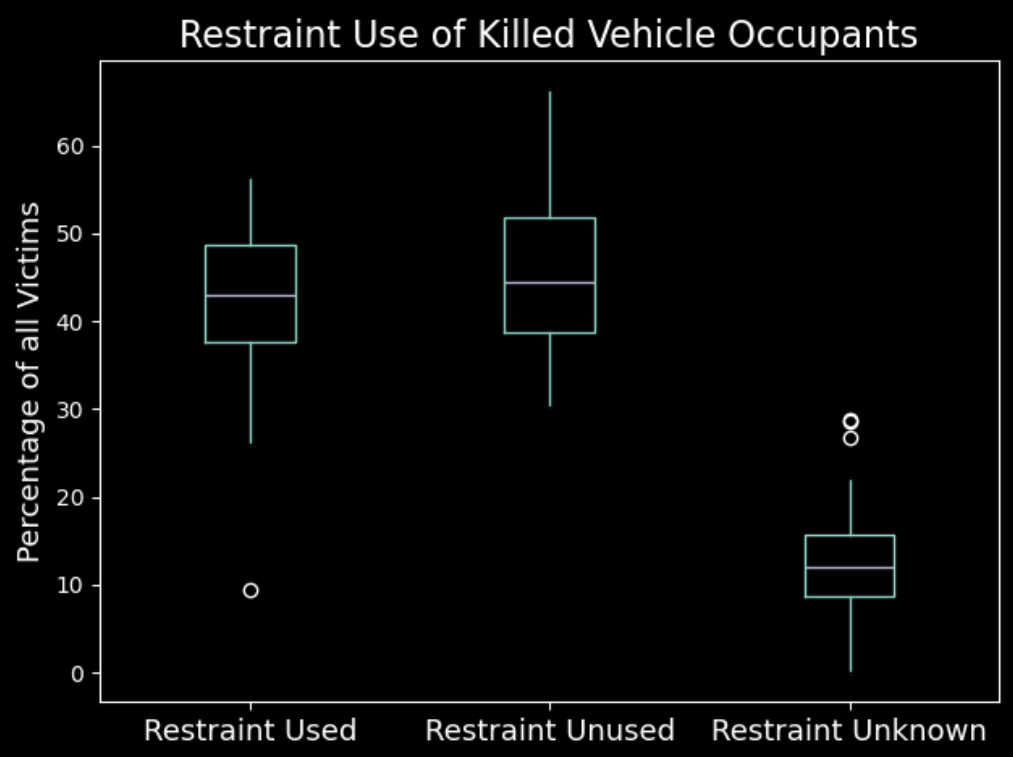
In a fatal crash:

- 20% more drunk driver died than survived
- In some states, the percentage is much worse
- **20% more sober drivers survived than died**



Restraint Use in Accident Fatalities

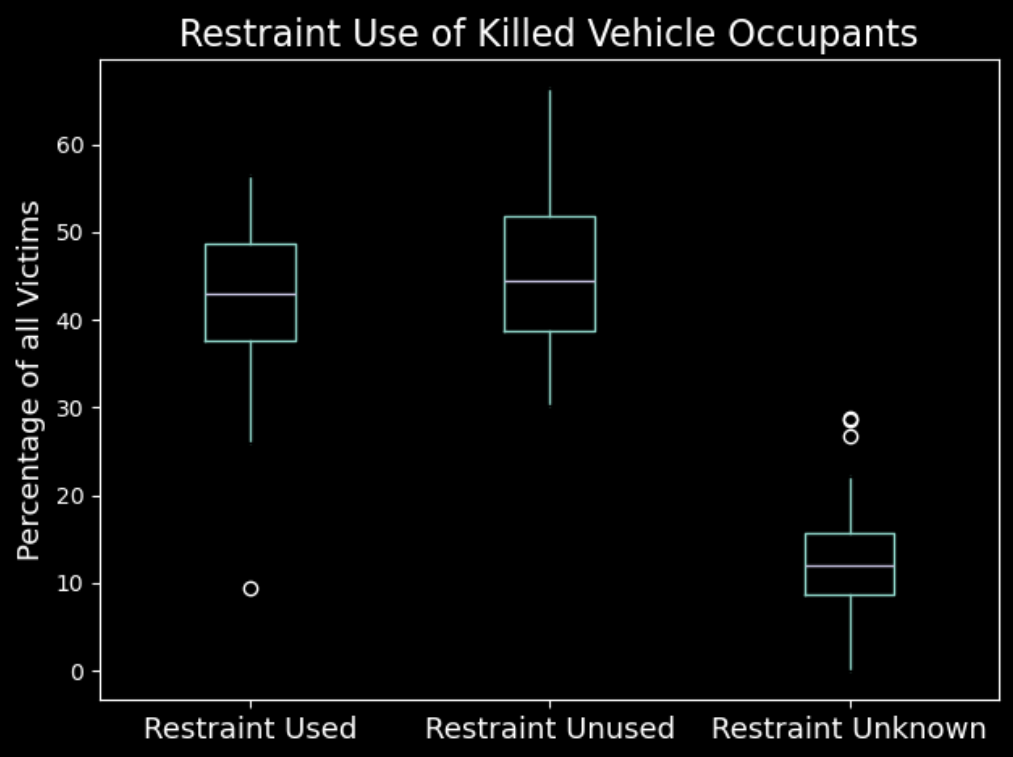
At first glance:



Restraint Use in Accident Fatalities

At first glance:

- **No big difference between restraints used and unused**



Restraint Use in Accident Fatalities

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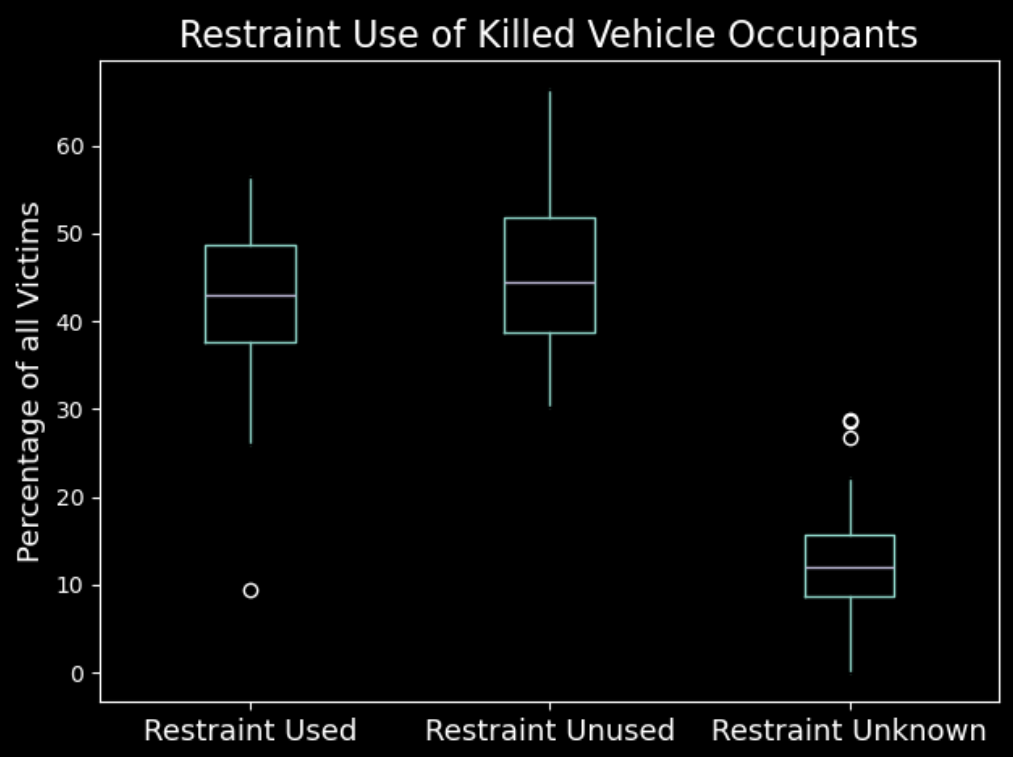
- No big difference between restraints used and unused
- **Much of the data is unknown**



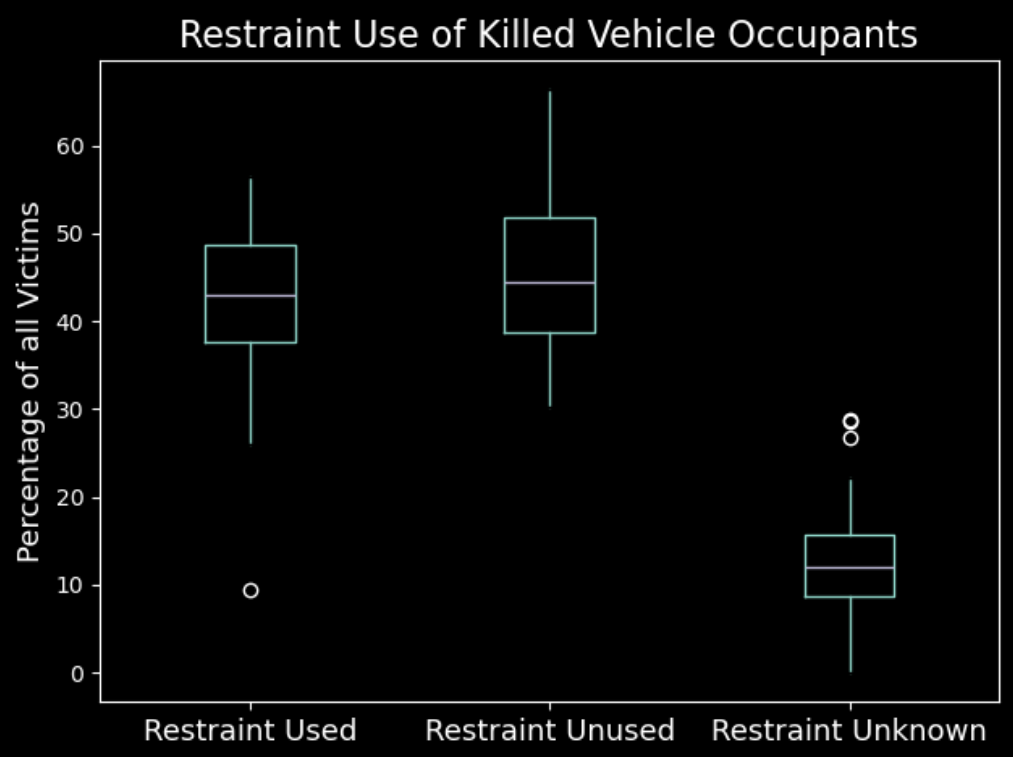
Restraint Use in Accident Fatalities

At first glance:

- No big difference between restraints used and unused
- Much of the data is unknown

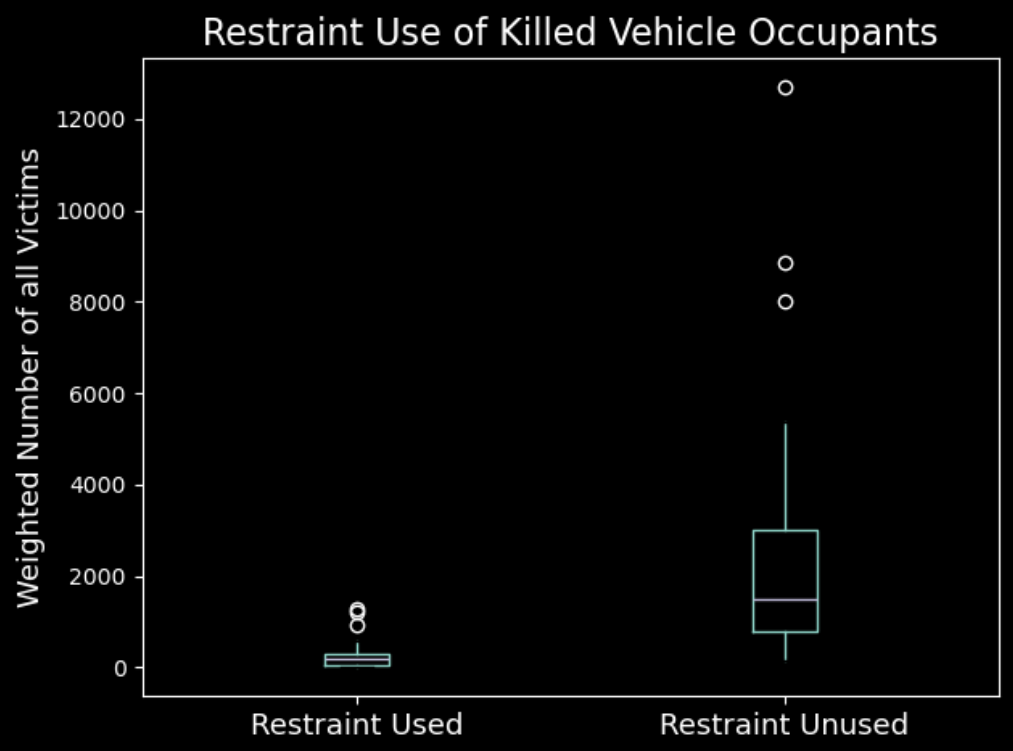
However:

- **Most people wear seatbelts**



Restraint Use in Accident Fatalities Continued

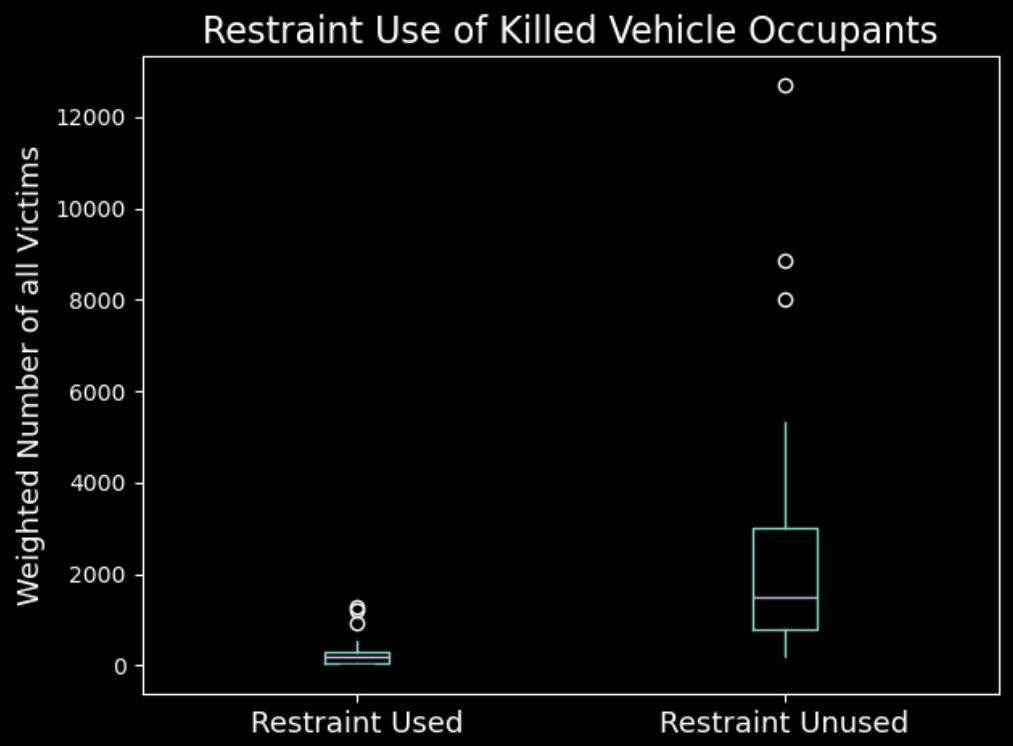
After weighting the data:



Restraint Use in Accident Fatalities Continued

After weighting the data:

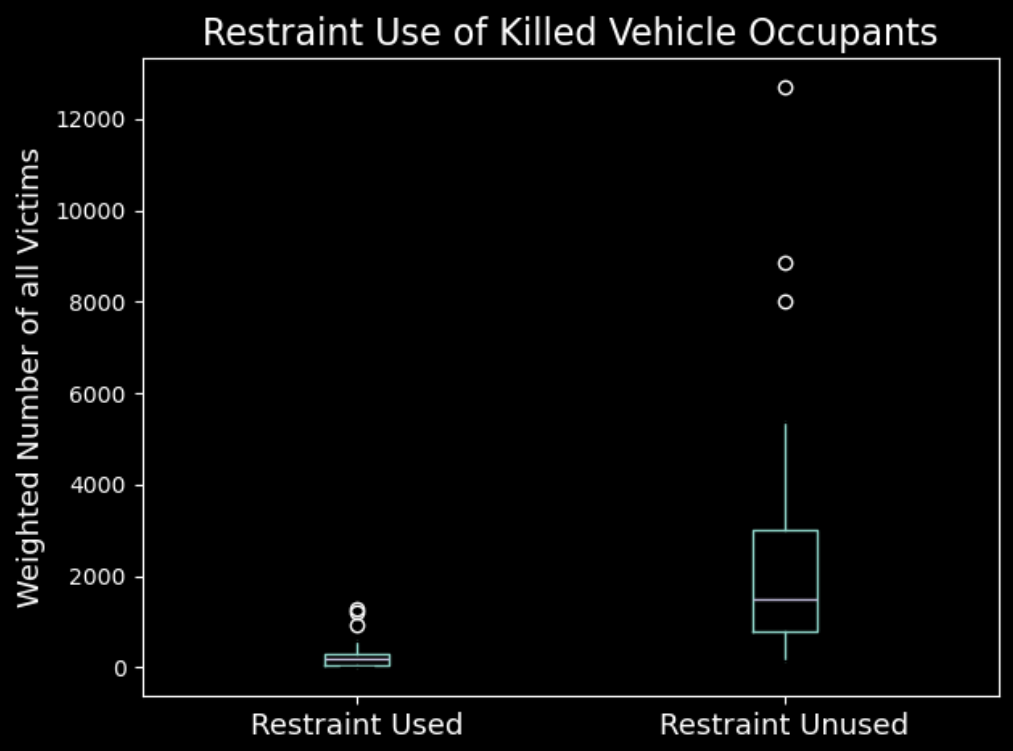
- **People without seatbelts are significantly more likely to die in a fatal crash**



Restraint Use in Accident Fatalities Continued

After weighting the data:

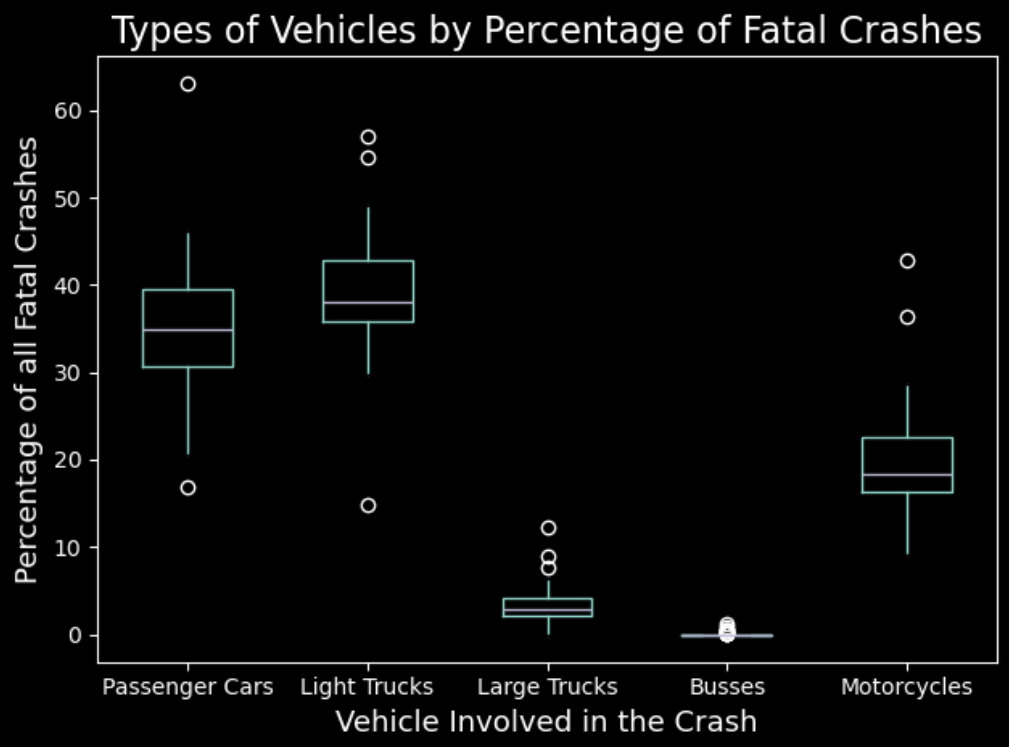
- People without seatbelts are significantly more likely to die in a fatal crash
- **Outliers represent high populations**



Percentage of all Fatal Crashes by Vehicle

At first glance:

- Light trucks/cars most dangerous
- Large trucks/busses least dangerous



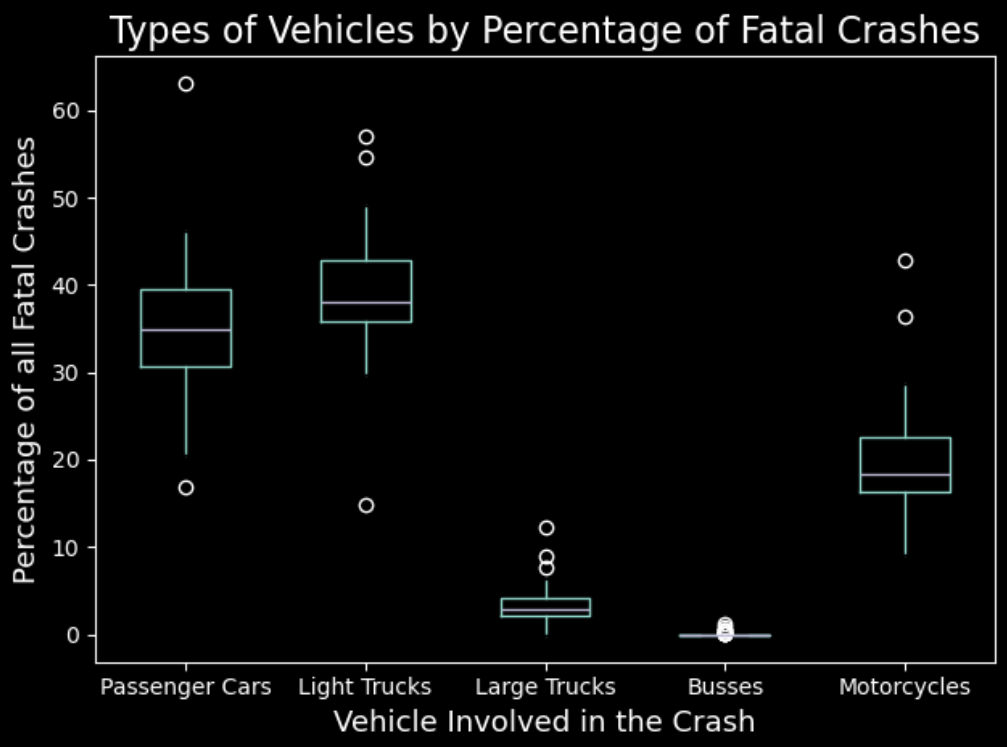
Percentage of all Fatal Crashes by Vehicle

At first glance:

- Light trucks/cars most dangerous
- Large trucks/busses least dangerous

However

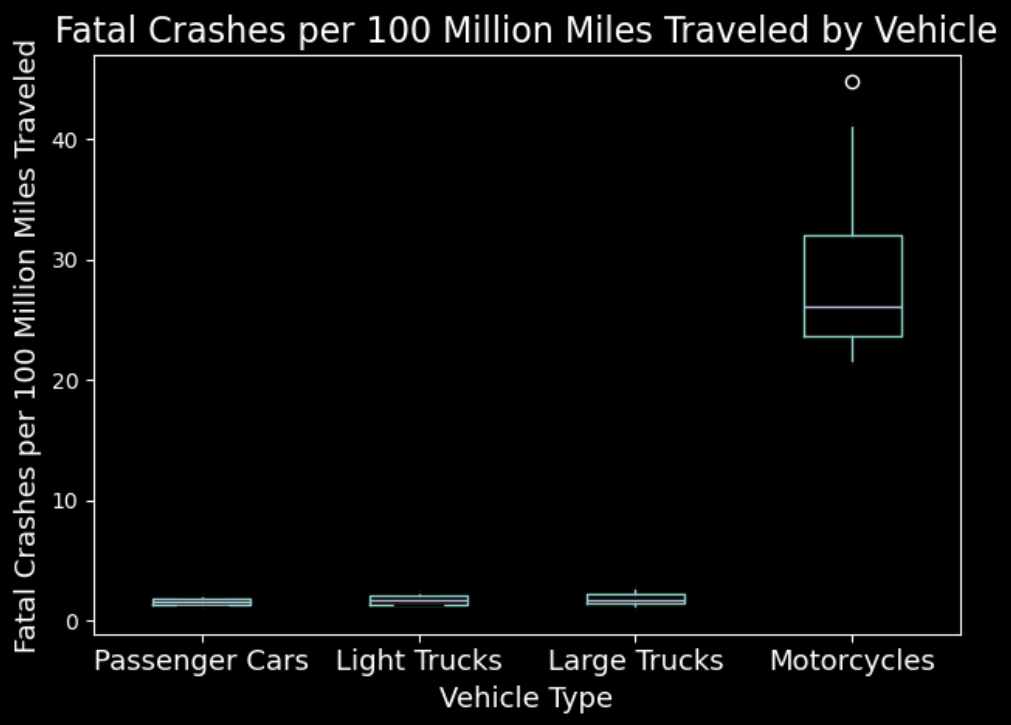
- **Those are most common vehicles**



Fatal Crashes per Million Miles Traveled by Vehicle

After weighting the data:

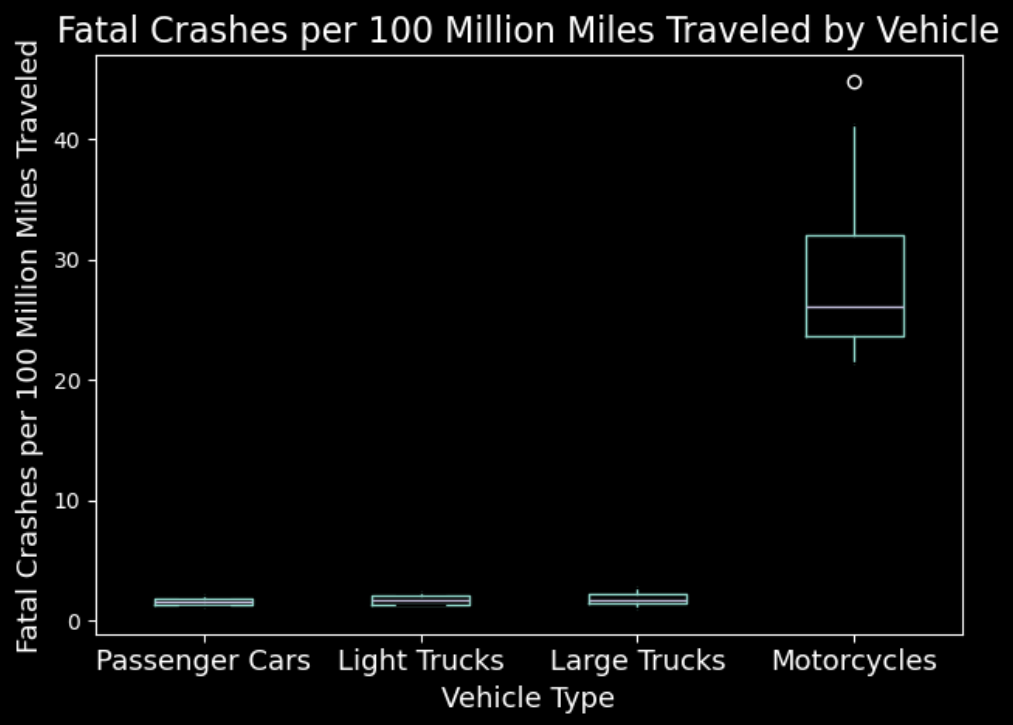
- **Motorcycles are the most dangerous**



Fatal Crashes per Million Miles Traveled by Vehicle

After weighting the data:

- Motorcycles are the most dangerous
- **Non motorcycles are near zero**



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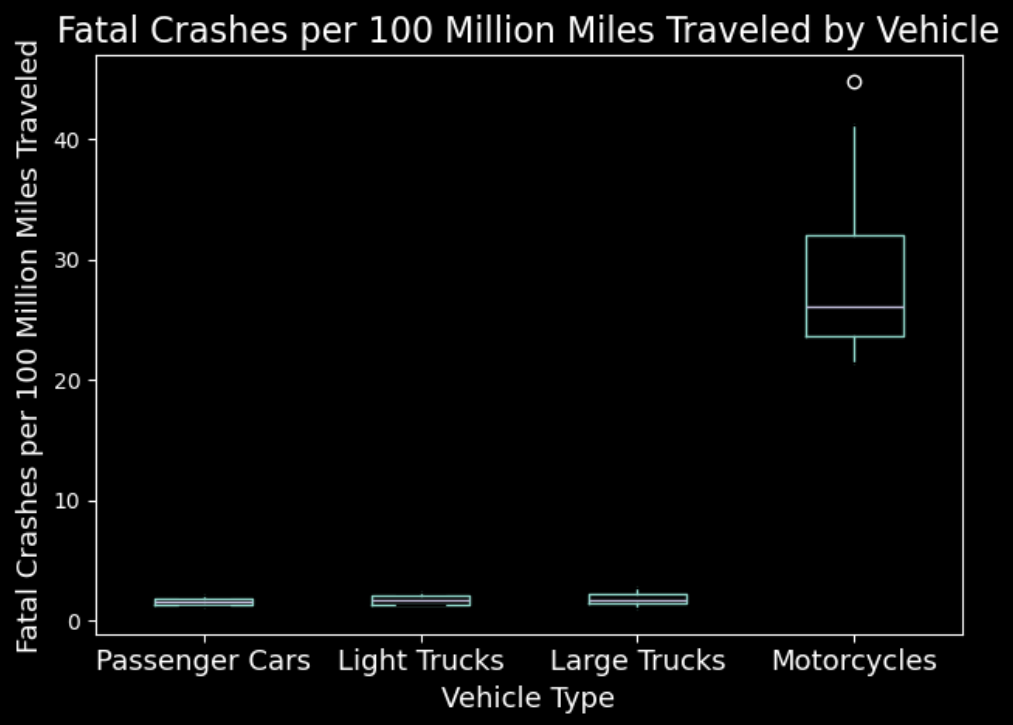
After weighting the data:

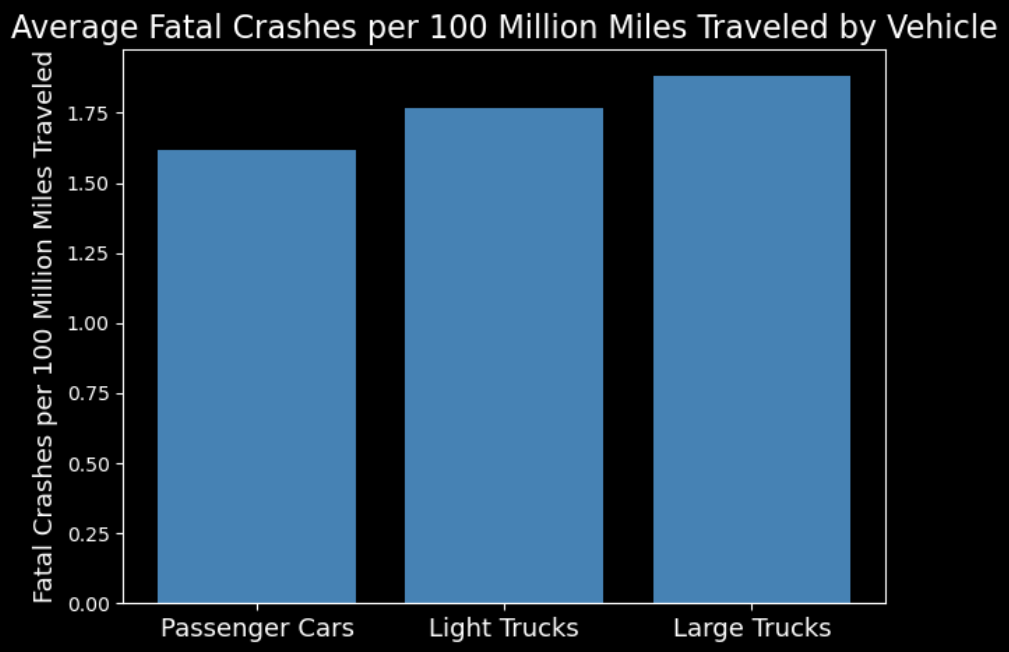
- Motorcycles are the most dangerous
- Non motorcycles are near zero
- **2005 was a bad year for motorcyclists**



Fatal Crashes per Million Miles Traveled by Vehicle

Excluding motorcycles:

- Large trucks are most likely to be in a fatal crash
- Passenger cars are least likely



Conclusions

- **Don't drink and drive**

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- **Wear your seatbelt**

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- Don't drink and drive
- Wear your seatbelt
- **Be cautious of motorcycles**

Future Directions

- **Examine accidents on a state by state basis**

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- **Investigate how this changes over time**

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- Examine accidents on a state by state basis
- Investigate how this changes over time
- **Consider factors causing these discrepancies**

References

- National Highway Traffic Safety Administration: FARS Encyclopedia: States - Occupants. (n.d.). Www-Fars.nhtsa.dot.gov. https://www-fars.nhtsa.dot.gov/States/StatesOccupants.aspx
- Pandas Assistance: Python Pandas: How to Convert Entire DataFrame to Numeric (int or float) — Tutorial Reference. (2026). Tutorialreference.com. https://tutorialreference.com/python/examples/faq/pandas-how-to-convert-entire-dataframe-to-numeric-integers-or-floats

Thanks for Listening!